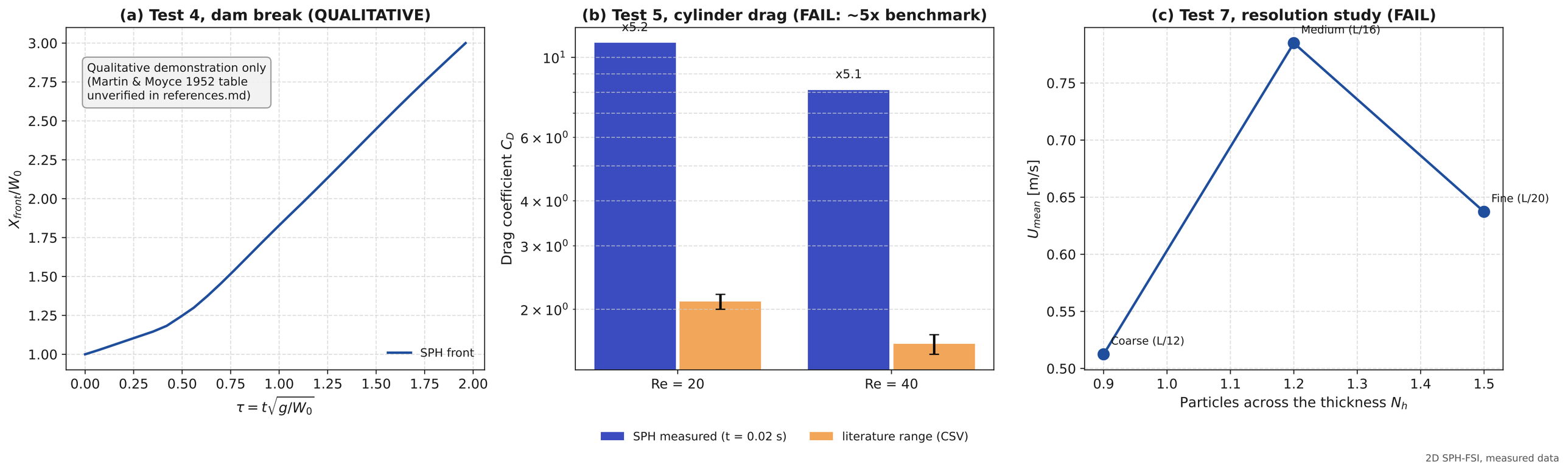
(a) Test 4, dam break (QUALITATIVE)
Qualitative demonstration only
(Martin & Moyce 1952 table
unverified in references.md)
SPH front
X_{front}/W_0
$\tau = t\sqrt{g/W_0}$
3.00
2.75
2.50
2.25
2.00
1.75
1.50
1.25
1.00
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
(b) Test 5, cylinder drag (FAIL: ~5x benchmark)
Drag coefficient C_D
10^1
6×10^0
4×10^0
3×10^0
2×10^0
x5.2
x5.1
Re = 20
Re = 40
SPH measured (t = 0.02 s)
literature range (CSV)
(c) Test 7, resolution study (FAIL)
U_{mean} [m/s]
0.75
0.70
0.65
0.60
0.55
0.50
Medium (L/16)
Fine (L/20)
Coarse (L/12)
0.9
1.0
1.1
1.2
1.3
1.4
1.5
Particles across the thickness N_h
2D SPH-FSI, measured data